EXPERIMENT 3

Multi-Byte Data and Array Operations Using 8086 Indirect Addressing
a) To add two 32-bit (double-precision) numbers using register-indirect and based addressing with SI, DI and BX.
b) To find the sum of N 16-bit numbers stored in a word array with a 32-bit result using a loop.

Objective:

To develop and verify **multi-byte data and array operations using 8086 indirect addressing**, including the addition of two **32-bit double-precision numbers** and finding the sum of **N 16-bit numbers** stored in a word array with a **32-bit result**.

Algorithm:

- Initialize the required data and memory locations.
- Use **SI, DI, and BX** registers for indirect and based addressing.
- Load the lower and higher words of the two 32-bit numbers.
- Add the lower 16-bit words and store the result.
- Add the higher 16-bit words along with the carry using ADC.
- Store the complete 32-bit addition result.
- For the array operation, initialize the count **N** and set the array pointer.
- Load each 16-bit number from the word array using indirect addressing.
- Add each number to the accumulator and maintain the result as a **32-bit value**.
- Repeat the process using a loop until all N numbers are added.
- Store and verify the final 32-bit result.

1)Addition of Two 32-bit numbers:

Code:

```
MOV SI,1000H
MOV DI,2010H
MOV BX,3000H
MOV AX,[SI]
ADD AX,[DI]
MOV [BX],AX
MOV AX,[SI+2]
ADC AX,[DI+2]
```

MOV [BX+2],AX

HLT

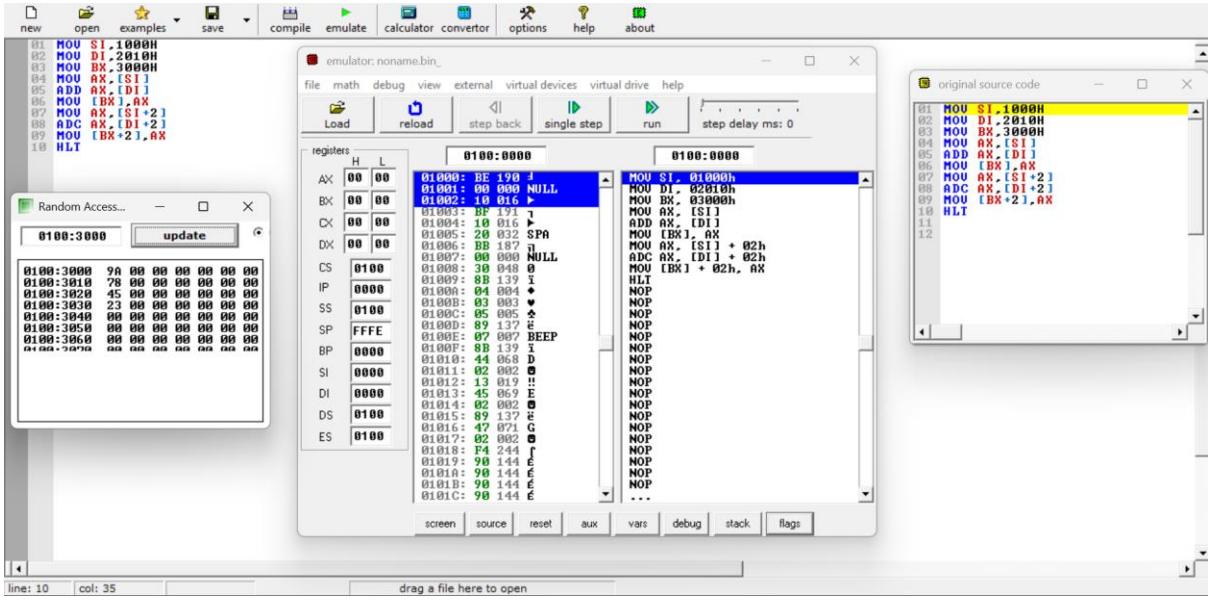
Input:

Input Address	Data
2000h	12345678h
2010h	11112222h

Output:

Output Address	Data
3000h	2345789Ah
3002h	0000h

Output Screenshort:



2)Sum of N 16-bit numbers:

Code:

MOV SI,2000H
MOV CX,0004H
XOR AX,AX
XOR DX,DX
LOOP1:
ADD AX,[SI]
ADC DX,0000H
ADD SI,0002H
LOOP LOOP1
MOV [3000H],AX
MOV [3002H],DX

HLT

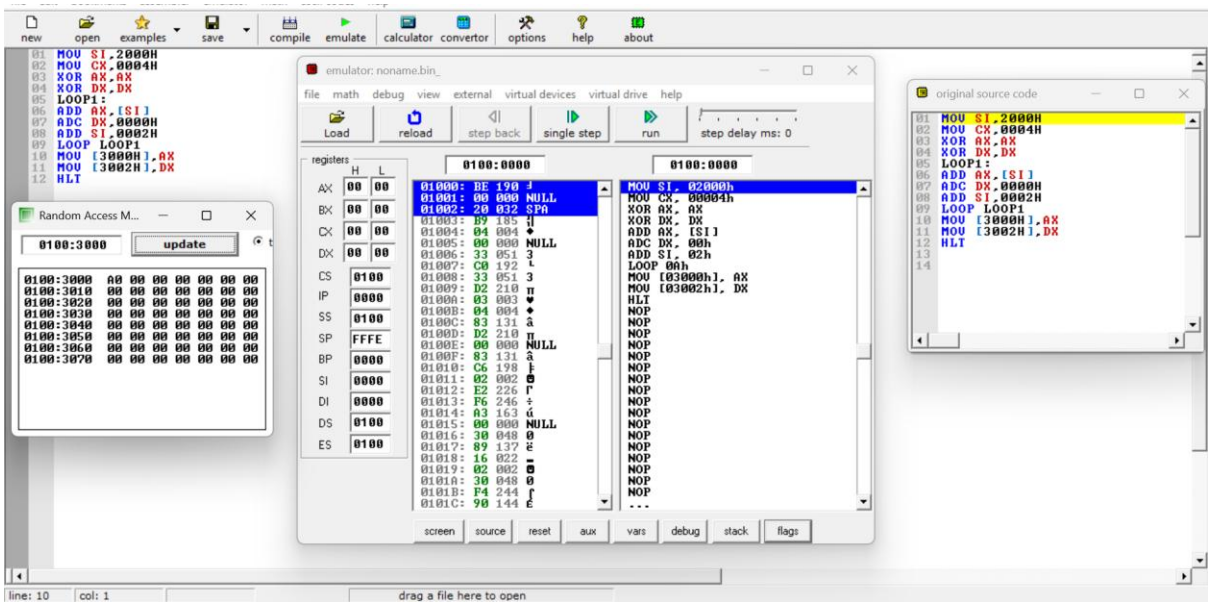
Input:

Input Address	Data
2000h	0010h
2002h	0020h
2004h	0030h
2006h	0040h

Output:

Output Address	Data
3000h	00A0h
3002h	0000h
-	-
-	-

Screenshort:



Result:

Thus, the addition of two **32-bit double-precision numbers** using **SI, DI, and BX indirect/based addressing** and the sum of **N 16-bit numbers stored in a word array** with a **32-bit result** were successfully performed and verified using the **8086 microprocessor**.

Student Reflection:

I learned how to handle **multi-byte data and arrays using 8086 indirect addressing**. I understood the use of **SI, DI, and BX registers**, carry propagation using ADC, and looping

through a word array. This experiment improved my understanding of **32-bit arithmetic operations, memory addressing, and array processing in 8086 assembly language**.